

SHAWN WEI

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Education

University of Waterloo

Sept 2025 – Apr 2029

Honours Bachelor of Computer Science, Co-op

Waterloo, ON

- Awarded the **Richard and Elizabeth Madter Entrance Scholarship** as the **sole recipient** from the incoming Faculty of Mathematics cohort for 2025

Experience & Leadership

TD Bank

May 2026 – Aug 2026

Data Engineering / Marketing Analytics

Toronto, ON

- Engineered distributed **PySpark** (Python + SQL) pipelines on **Azure Databricks** to clean and transform relational data across **15M+** active Canadian customer profiles
- Built a **PySpark** notebook-generation tool that templates **SQL** and Python queries against the customer database, standardizing **400+** analytics notebooks and eliminating inconsistent manual querying
- Built **cross-reference validation** tooling to reconcile records across migrated datasets, catching data leaks that would have launched marketing campaigns on **inaccurate data**
- Engineered a **TD Prism**-powered **RAG assistant** over TD's internal **Salesforce CDC** database, giving **170+** colleagues self-serve natural-language access to **1,250+ marketing segments**; recognized by the **VP of Marketing Technology**

FRC Team 9569 Silver Knights

Oct 2022 – Jan 2025

Programming Lead

Markham, ON

- Orchestrated the software development lifecycle for an **8-person** programming team, enforcing **Git** branching strategies and Agile methodologies to ensure rapid code iteration during a 6-week build season
- Architected a modular **Java** codebase using the **WPILib** framework, reducing technical debt and improving code maintainability by **25%** through strict object-oriented design principles
- Programmed **autonomous routines** using **OpenCV** and **AprilTag** detection for real-time target acquisition and path planning, tuned with **PID control** for precise positioning
- Strategized software-hardware integration that secured the team's first-ever **Rookie All-Star Award**, achieving a top-40% provincial ranking (**53/138**) in the debut season

Skills

- **Languages:** Python, Java, C, C++, JavaScript, TypeScript, SQL, Bash
- **AI/ML & Data:** LLM fine-tuning (LoRA), FP8 quantization, PySpark, Pandas, NumPy, Databricks
- **Frameworks:** React.js, Next.js, Node.js, FastAPI, REST APIs, Flask, Django, TailwindCSS, WebSockets
- **Tools & Platforms:** Azure, Docker, Git, Linux/Unix, CI/CD, OpenCV, Salesforce

Projects

Ensemble – Finalist at Hack the 6ix

Jul 2026

Python (asyncio), JavaScript, C++ | WebSockets, OpenCV, Qwen3.5 (LoRA), Fireworks, Arduino

- Demoed live to **230+ hackers** and sponsors, turning **200+ audience phones** into one gesture-conducted orchestra with **synchronized** browser audio via a **Python asyncio + WebSocket** server using NTP-style clock sync
- Fine-tuned **two Qwen3.5 LoRA** models that compose accompaniment live each bar, merged to **FP8** and served on a dedicated **Fireworks H200** at **453 ms/bar** (10× faster than shared serving)
- Guaranteed **gapless** playback by prefetching accompaniment **2 bars (~4.8 s)** ahead and falling back to a deterministic engine on any missed request

Real-Time Flight Telemetry & Analytics Pipeline

Dec 2025

Python, JavaScript | Flask, SQLAlchemy, PostgreSQL, NumPy, Leaflet, OpenSky API

- Engineered a high-throughput ingestion pipeline to process real-time **ADS-B** signals, ensuring **data integrity** while transforming unstructured JSON telemetry into relational schemas
- Reduced query retrieval latency by **40%** for **50+** concurrent streams by implementing efficient **indexing strategies** and local caching
- Applied **statistical analysis** with **NumPy** to surface telemetry trends, providing actionable insights into aircraft performance metrics